

Lab #1:-

Immune System

Innate (Natural)

Adaptive (Required)

- non specific

- Specific

- not memory

- memory

- no need for expose to foreign substances.

- need for expose to foreign substances

(e.g.: tears, skin, gastric juice, macrophages).

e.g.: T-cells, B-cells

↓
cellular↓
humoral

* Antigen (Ag) → foreign substance that could inhanse immune system or couldn't inhanse immune system.

* Immunogin → foreign substance that must inhanse immune system.

* Characteristics of Ag:-

1- Protein (high M.W.).

2- free of toxic.

3- Steriled (no M.O.).

4- Pure (one epitope).

5- suitable pH.

6- Dose. (*)

Lab #2 : Ag and Ab / Immunization

- * Dose :- amount of Ag per kg of animal.
- high dose :- as self Ag and tolerance (no immune response).

- low dose :- no immune response.

- Suitable dose → strong immune response → Ab

* Route of admission for Ag in to Body :-

- ① Nose + mouth
- ② Intradermis (I.D) → inside skin
- ③ Intra peritoneal (I.P) → inside peritoneal cavity.
- ④ Intra veinous (I.V) → inside veins.
- ⑤ Intra muscular (I.M) → Inside muscles.
- ⑥ Subcutaneous (C.S) → under skin.

* Animals :-

- ① Rat
- ② Mice
- ③ Rabbits
- ④ Chicken
- ⑤ Hamster

- Mice :- albino, white color why??

1- 98% genetically identical. مكتوبة متساوية

2- Interbreeding between sisters and brothers more than 20 times.

Marking of Animals

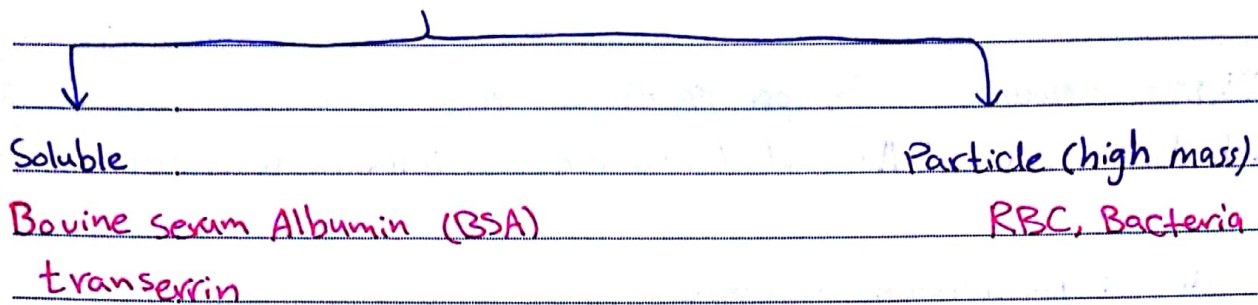
Permanent

- long time
- large animals :- cow, sheep
- breeding or burning
- small animals : ear banding (mice, rabbits, rat)

temporary

(short time) add stain/
add label.

* Types of Antigen :-

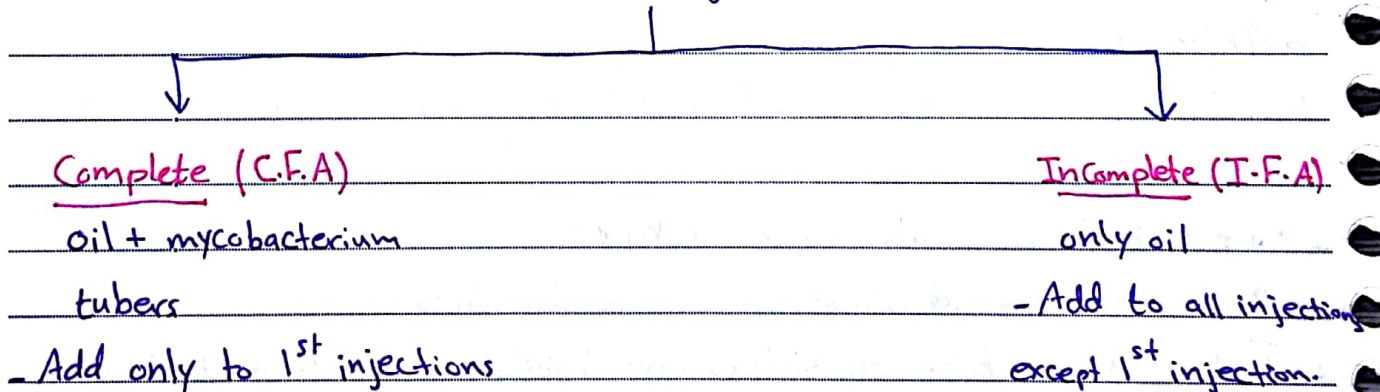


$Ag + Ab \rightarrow ppt$ راسب

* Adjuvant :- substance add to only soluble Ag to all route of admission except I.V to do the following

- 1- slow release Ag to tissue.
 - 2- slow absorb Ag by tissue.
 - 3- Enhance Immune system induction.
- } to prevent check for immune system

Frud Adjuvant



* Ag Bovine serum Albumin (soluble protein) with C.F.A

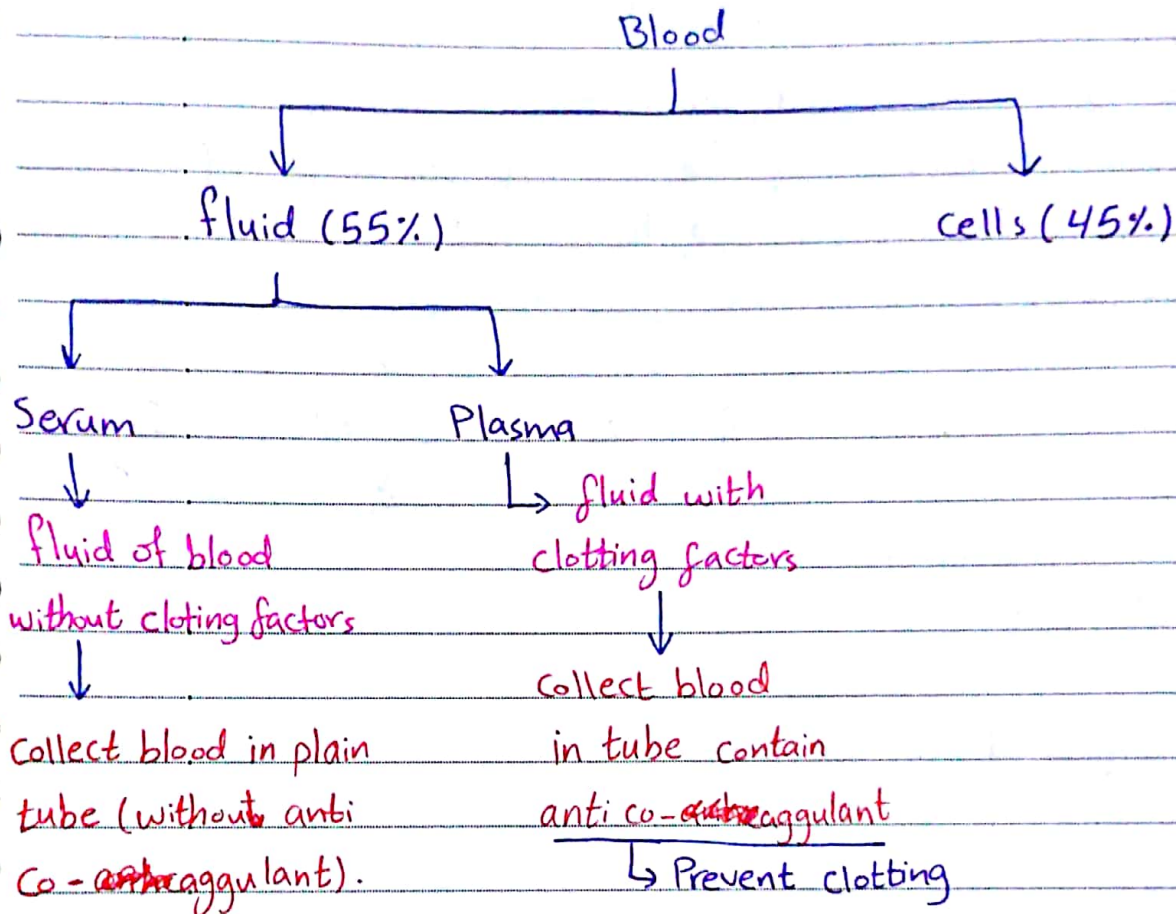
BSA	CFA
1	1

100 μ L : 100 μ L $\Rightarrow$ 200 mL / 1 mouse

Dose of BSA / mouse 50 mg - 100 μ g
route of admission I.P.

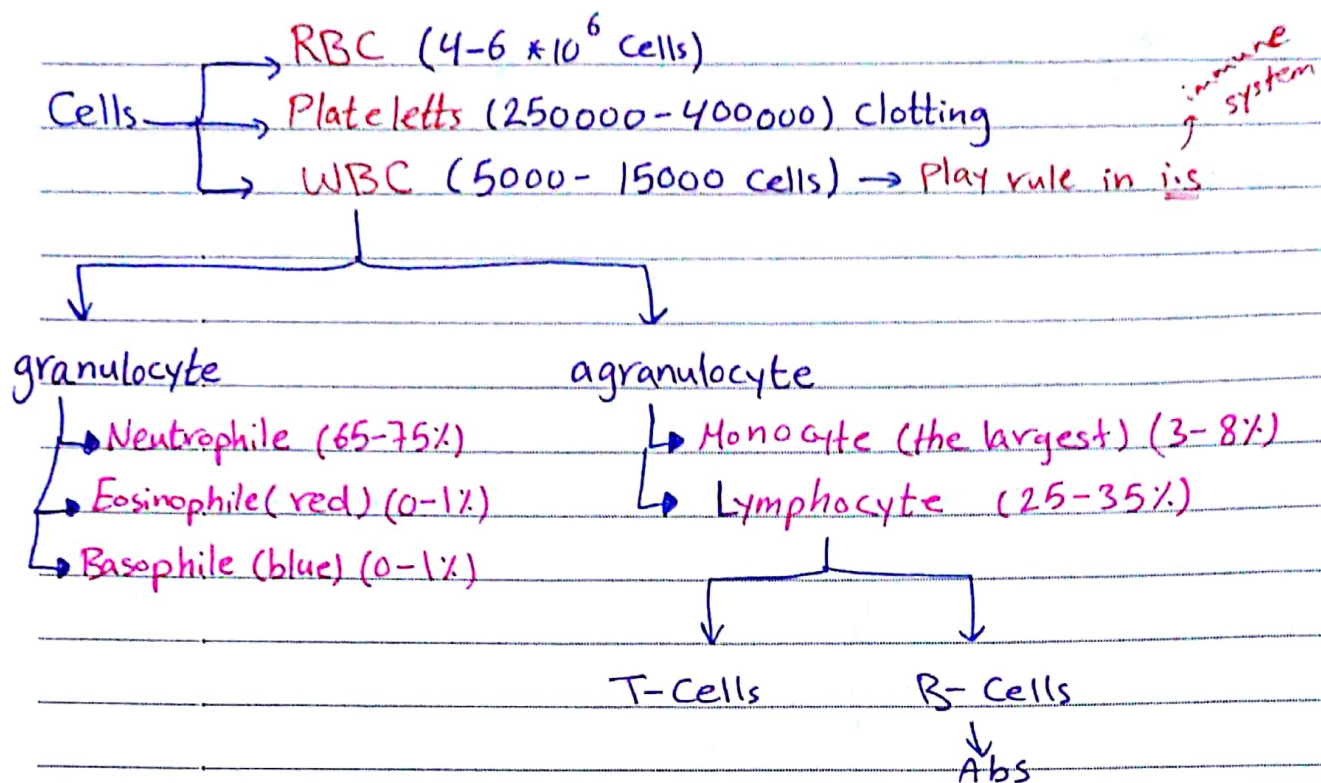
Lab #3:- Blood

* Blood $\rightarrow$ type of connective tissue.



* 4 types of co-~~ag~~agglulant:-

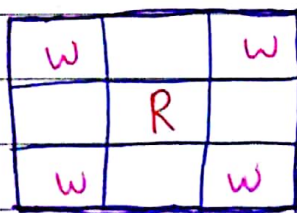
- 1- EDTA
- 2- oxalate $\rightarrow$ attack Ca^{+2}
- 3- Citrate
- 4- Heparin $\rightarrow$ in tissue culture



* Count of WBC :-

Neuber improved Hemocytometer

↓ blood ↓ cell

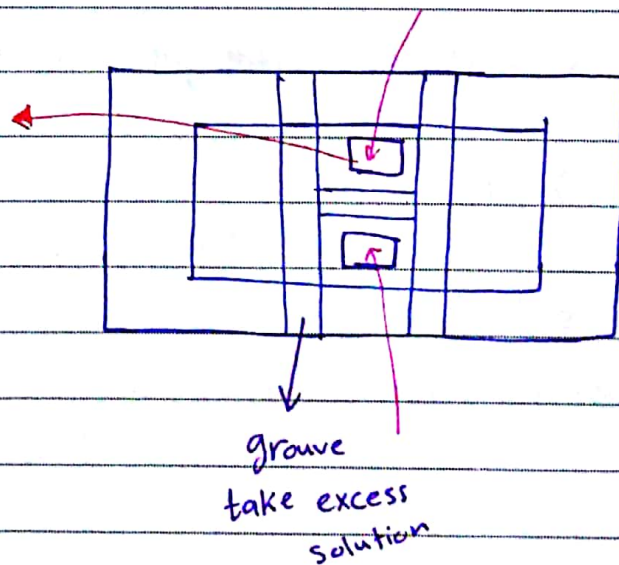


length = 1mm

width = 1mm

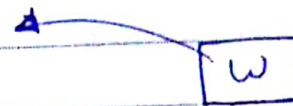
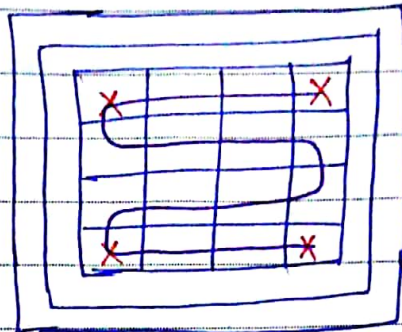
depth = 0.1mm

Volume $W = 0.1 \text{ mm}^3$



W $\Rightarrow$ Count large cells (WBC, macrophage, Fungus, Epith).

R $\Rightarrow$ " small cells (RBC, yeast, bacteria, sperm cells).



16 small squares

Count at any corner start = zig zag lines

-2/4 $\rightarrow$ minimizing errors (more details in Hematology Lab).

$$\text{Calculation} = \# \text{ of cells in } W * \frac{V.CF}{\frac{1}{0.1}} * \frac{DF}{20} = \text{--- cell/mm}^3$$

$$V.CF = \text{volume} * \text{conc. factor} = \frac{1}{\#W} = \frac{1}{0.1}, \frac{1}{0.2}, \dots$$

$$DF = \frac{\text{Final } V}{\text{Sample } V} = \frac{950 \mu\text{m (acetic acid)} * 50 \mu\text{m sample}}{50 \mu\text{L}} \\ = \frac{1000}{50} = 20$$

* function of acetic acid $\rightarrow$ lysing RBC.

Procedure :-

1- Sample acetic acid

$\Rightarrow$ white blood (in agglutinant) mixing until brown color of soln appear. (lysing).

2- Prepare hemocytometer, clean it, clean microscope and put cover slip on hemocytometer.

3- Put hemocytometer on microscope at low object (4X)

4- No apply sample by pipette or capillaries.

5- Count WBC $\rightarrow$ shine & dark at medium (10X)

6- Calculations.

* Blood Smear (film):-

1- See shape of RBC, volume, color

2- See some paracyte.

3- " platelets.

4- Count for different shape of WBC (100 cells).

* Procedure:-

1- Take whole blood (with anticoagulant), put few drops on slide.

2- Spread blood on slide tongue shape, leave for air dry.

3- Add stain $\rightarrow$ for 1 min (right stain).

4- Wash by tap water, leave dry.

5- See under microscope (40X)

* Right stain:-

mix:-

① Acidic stain $\rightarrow$ red Eosin stain

② Basic stain $\rightarrow$ blue hematoxylin

Fibrinogen $\xrightarrow[\text{enzyme}]{\text{Ca}^{+2}}$ Fibrin

↓ ↓

inactive liquid active thread
 solid

→ Clothing

15-30 min.

①, During Survival (still alive) → - small volume

- need animal for other researches.

b- Cardiac puncture from heart.

c- Vein puncture → Tail (mouse)
 ↳ ear (rabbit)

-need high amount of blood.

- no need for animals.

① Coagulation clotting (blood in plane tube).

② Centrifugation at 3000 rpm for 5 min, transfer serum to ependruff tube.

③ Cleaving:- clear serum by remove fat

fasting animal 12 hrs

→ Ultra centrifuge 25000rpm
15min at 4°C

④ Preservation: adding of preservative → to stop cell division of H.O

- Methiolate
- Phenol
- Sodium azide.

⑤ Storing :-

- Keep for long time at -20°C
- need IgG at -70°C
- Powder serum at 4°C

} → lypolization

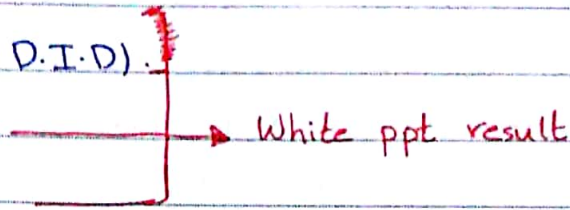
↳ Serum → other purification (ammonium sulfate).

Lab #5 :- Precipitation Test (ppt Test).

1- Double Immune Diffusion (D.I.D).

2- S.R.I.D

3- I.E



* BSA (Soluble Ag) :-

Soluble Ag + Soluble Ab $\rightleftharpoons$ ppt test

small size of Ag

Large particles Ag + Soluble Ab $\rightleftharpoons$ agglutination.

* Factors affect Ag-Ab complex:-

1- Conc. of Ag, Ab ($[Ag]$, $[Ab]$).

2- Physical factors (pH, temperature, pressure).

3- Affinity of Ab (strength between Ab and Ag).

* Rxn between Ag and Ab zones :-

1- Pre-Zone :-

$$[Ag] \ll [Ab]$$

- There is no white ppt result

- few ppt but not visible.

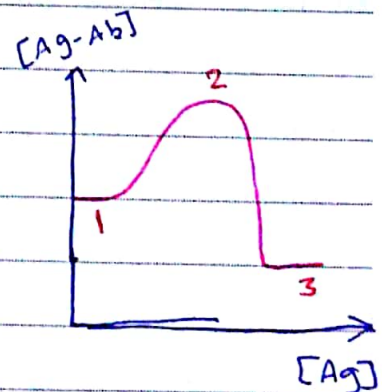
2- Eq-Zone :-

$$[Ag] \approx [Ab]$$

There is white ppt result (visible).

3- Post-Zone :-

$[Ag] \gg [Ab] \Rightarrow$ There is no white ppt result.



*D.I.D (Auchterlony Test):-

D → Double → Ag and Ab move to each other.

I → Immune → Ag-Ab rxn.

D → Diffusion → Principle of techniques (moving from high conc. to low conc.).

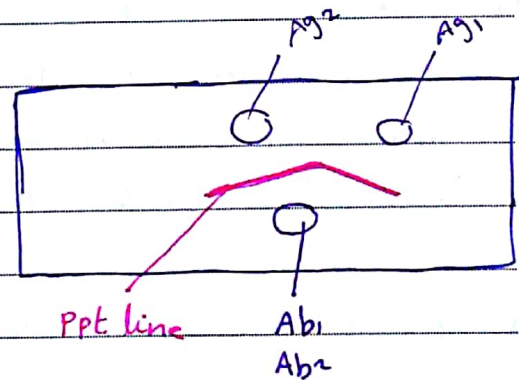
Applications:-

- 1- Test for homogeneity between Ag and Ab.
- 2- Test for purity of Ag (serum).
- 3- Test for semiquantitative (comparison).

(1+2) ⇒ determination by rxn of identity:-

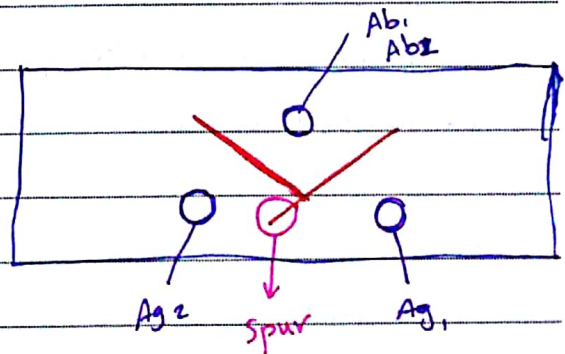
• Identical rxn:-

⇒ If white ppt lines are fused together, so Ag is identical to Ag₂.



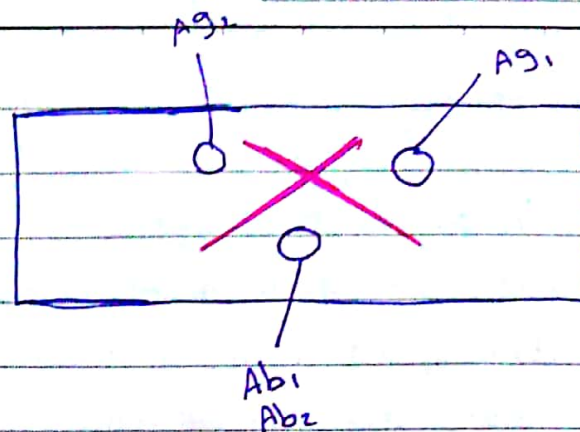
• Partially identical rxn:-

⇒ If white ppt lines are ~~crossed~~ fused together with form, so Ag₁ is partially identical to Ag₂.



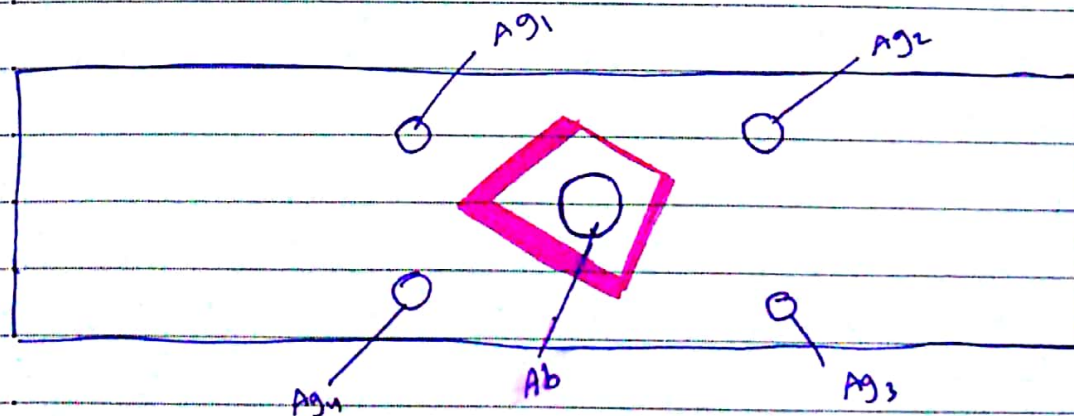
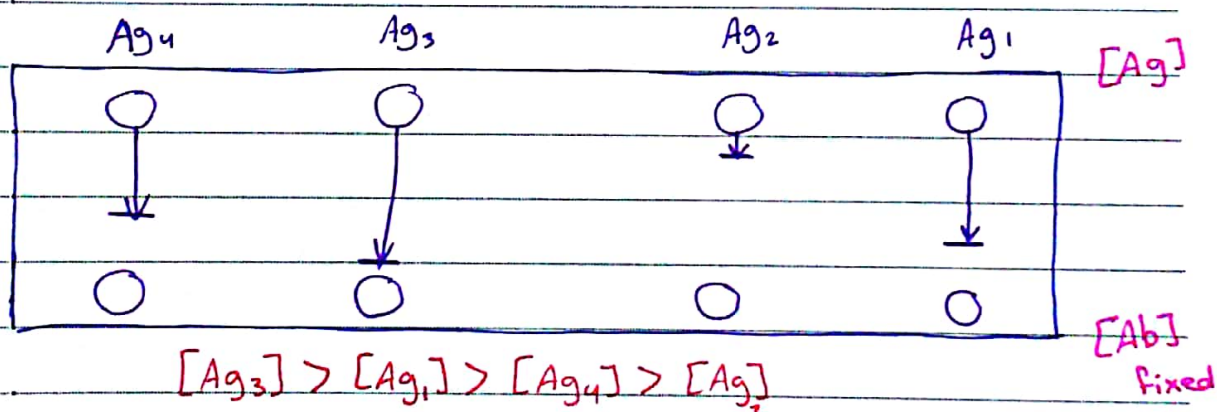
• Non Identical rxn:

⇒ If white ppt lines are crossed,
 Ag_1 is not identical to Ag_2 .



Semiquantitative :-

- * Distance travel by Ag or Ab ↑, $[Ag]$ ↑ or $[Ab]$ ↑
- * Thickness of ppt line ↑, $[]$ ↑

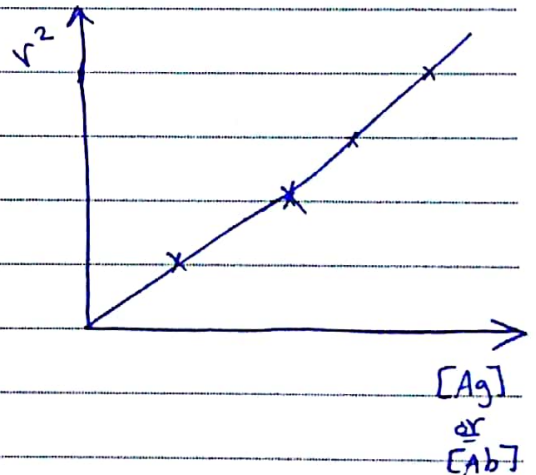
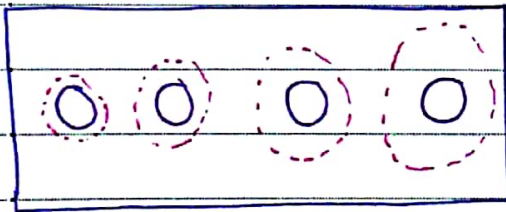


Working:-

- 1- Precoating of 1% agarose.
- 2- Add 5-6 ml melting agarose at (53°C), 2% → leave for 15-30 min (to be solid).
- 3- Add adaptor on surface of agarose (to make wells).
- 4- Remove agar from ~~well~~ wells.
- 5- Add 10 µL anti mouse IgG (Ab).
- 6- Add 10 µL mice serum around.
- 7- Incubate slide at humid chamber (at room temp.) for 24-48 hrs.
- 8- See your result.

Lab #6 :- SRID (Mannsin) :-S $\rightarrow$ Single (one of them: Ag or Ab).R $\rightarrow$ Radius $\rightarrow$ ppt hallow circle radius.I $\rightarrow$ Immuno $\rightarrow$ Ag-Ab rxnD $\rightarrow$ Diffusion (higher ions).* Applications :-

- Measure:- Quantitative for [Ag] or [Ab].

diameter $\uparrow$, [Ab] $\uparrow$ or [Ag] _{unknown}* Procedure :-

1- Pre coating slide with 1% agarose.

2- Add 100 μ L Anti mice with 5-6 mL melting agarose

or " " Anti human " " " " "

52-53°C

3- leave slide for 15-30 min to be solid.

4- Serul dilution for human serum and mice serum

 $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$

Make

- 5- ~~Add~~ 5 wells on agar and remove agar from wells.
- 6- Add 10 μ L from each dilution in each well in ascending or descending.
- 7- Put slide in humid chamber.
- 8- Come to see slides after 24-48 hrs.

Lab # 7 :- I.E

white ppt in I.E

* Ag is soluble.

I → Immuno (Ag-Ab rxn).

E → Electrophoresis → separation of substance by electricity due to charge.

* I.E has 2 methods:-

1- Electrophoresis.

2- D.I.D.

* Applications:-

1- Separation of Ag or serum mixture.

2- Due to charge.

3- Semiquantitative for immunoglobulin

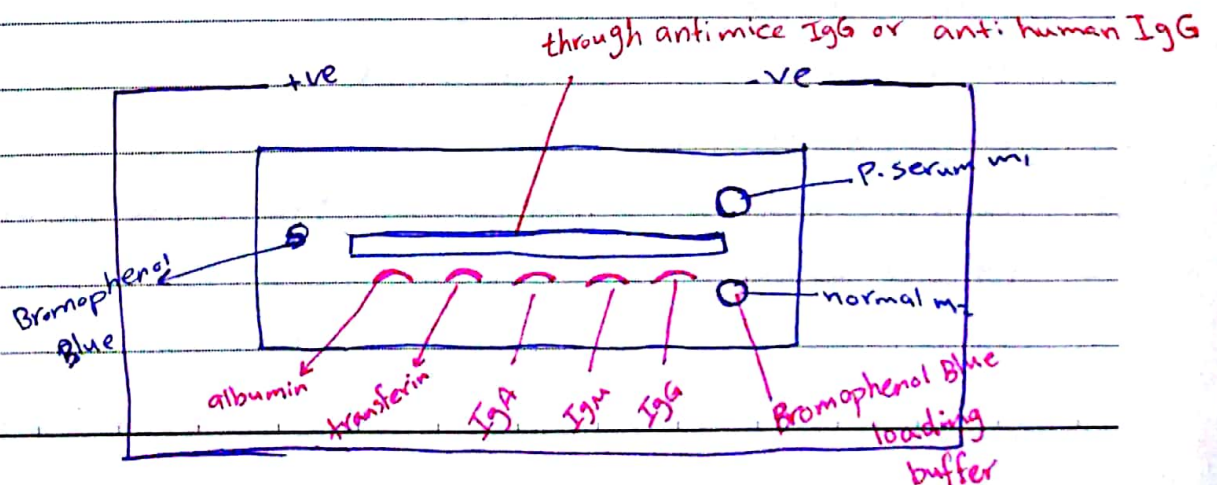
IgG
IgM
IgA
IgD
IgE

* I.E has advantages over D.I.D:-

1- Less chance for over lapping.

2- separated of mixture ~~due to~~ due to charge.

3- used for clinical significance related to blood or Igs.



* tris-HCl $\Rightarrow$ pH = 8.6, cover 90% of proteins running for e^- s with -ve charge so samples add near -ve and move to +ve (Some proteins have zero or +ve charge so left one finger space).

* Loading buffer $\rightarrow$ glycerol + bromophenol blue.

(density of sample and tracing of electrophoresis).

T = 35-40 min

V = 120V

* Albumin $\rightarrow$ high negative charge.

* If one white ppt arc appears $\rightarrow$ Homogenous Abs (one type)

* If two or more than one white ppt arc appear $\rightarrow$ Heterogenous (2 or more types)

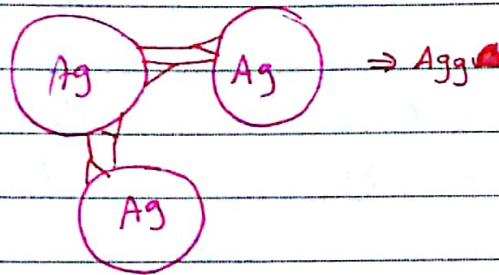
Procedures:

- 1- Precoating slide with 1% agarose.
- 2- 5-6 ml melting ~~ag~~ 2% agarose.
- 3- leave slide 15-30 min to be solid.
- 4- Make two wells and between them (left one finger space).
- 5- Add 10 μ L in each well from different sample.
- 6- Add 10 μ L of loading buffer on one sample.
- 7- Put slides in electrophoresis at -ve electrons.
- 8- Add tris-HCl pH = 8.6 to counter.
- 9- Run electrophoresis for 35-40 min at 120V.
- 10- Remove over from through.
- 11- Incubate slides at humid chamber (at room temp. for 24-~~34~~ 48 hrs).
- 12- See your result.

Lab #8 :- Agglutination (Serology)

* ppt $\rightarrow$ Soluble Ag + Soluble Ab

* Agglutination $\rightarrow$ clumping between Particle Ag with soluble Ab and give visible rxn at equivalent zone.

* Types of Agg:-

1- Direct agg $\rightarrow$ Ag-Ab rxn, no intermediate substance whether looking for Ag or Ab.

2- Passive agg (indirect) $\rightarrow$ we have intermediate substance between Ag & Ab.

Subtypes (linkage):-

1- latex (inert substance).

• Direct latex agg $\rightarrow$ looking for Ag
 $\rightarrow$ latex bind Ab

• Indirect latex agg $\rightarrow$ looking for Ab
 $\rightarrow$ latex bind Ag

latex $\rightarrow$ white color

2- Hem. (RBCs)

• Direct hem agg $\rightarrow$ looking for Ag
 $\rightarrow$ bind Ab

• Indirect hem agg $\rightarrow$ looking for Ab
 $\rightarrow$ bind Ag

مستفاد من

*Direct Agg → color rxn, the background is white).

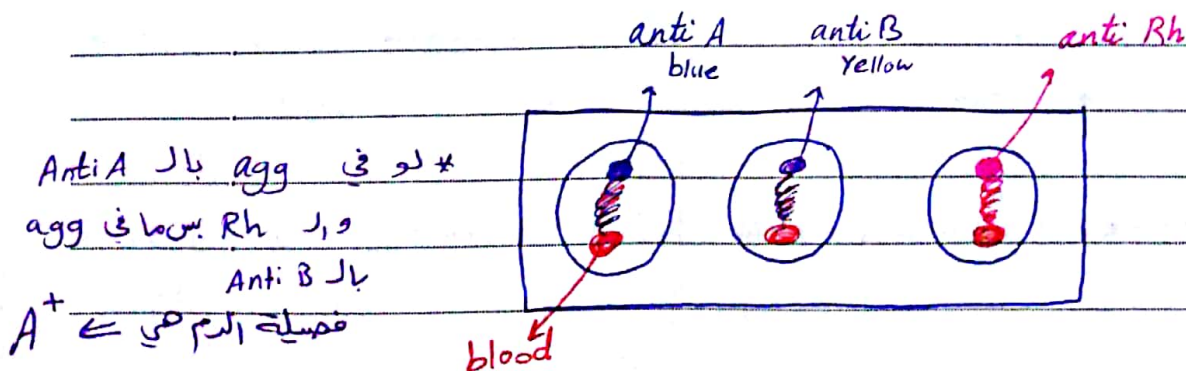
* Applications:-

1- Blood transfusion → ABO system

2- Bacterial types.

3- Ab titer ([]).

Blood type Phenotype	Ag Present on surface of RBC	Ab Present fluid	Possible genotype
 A	Ag A	Ab B	$I^A I^A, I^A i$
 B	Ag B	Ab A	$I^B I^B, I^B i$
 AB	Ag A, Ag B	non	$I^A I^B$
 O	non	Ab A, Ab B	ii
 +ve	Rh	none	$Rh Rh, Rh rh$
 -ve	non	non	$rh rh$

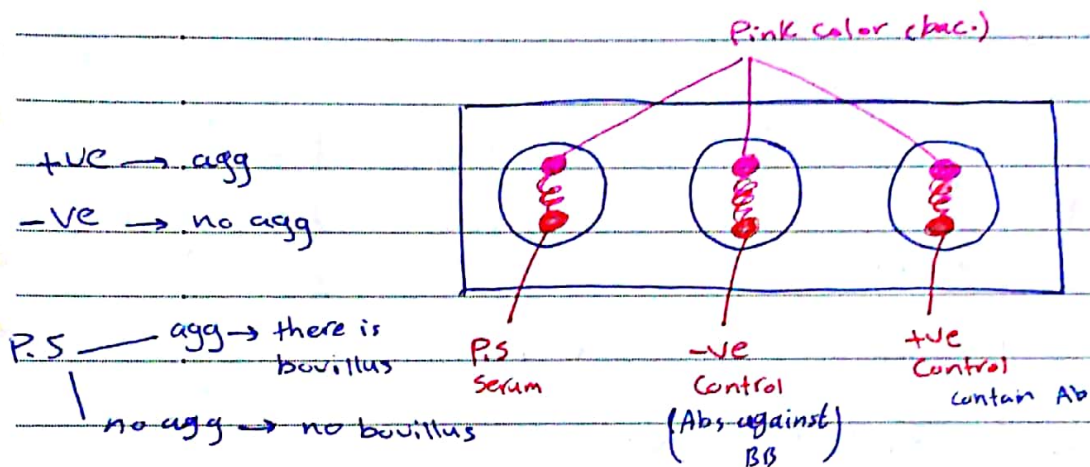


* Rose Bengal (Pink color)

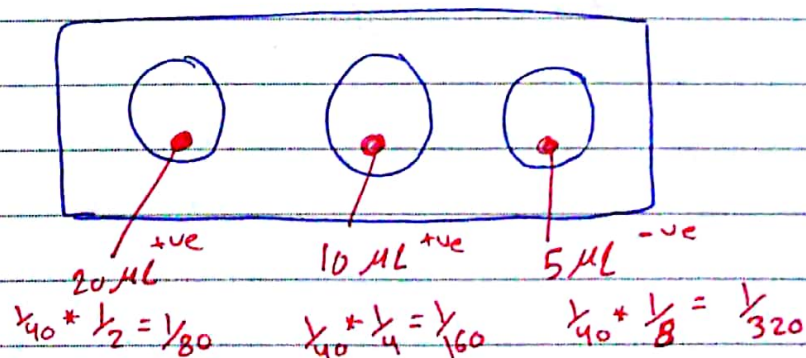
* Detect present of Ab in serum against *Brucella bovis* → this bacteria causes Brucellosis → قملل, سق

due to contaminated diary product with this bac. (cows, sheep)

Pink ⇒ weaken pink *brucella bovis*.



+ve P.S :-



Titer > 80 ⇒ infected

Titer < 80 ⇒ Previous infection

Titer = 80 ⇒ repeat test after 1 month

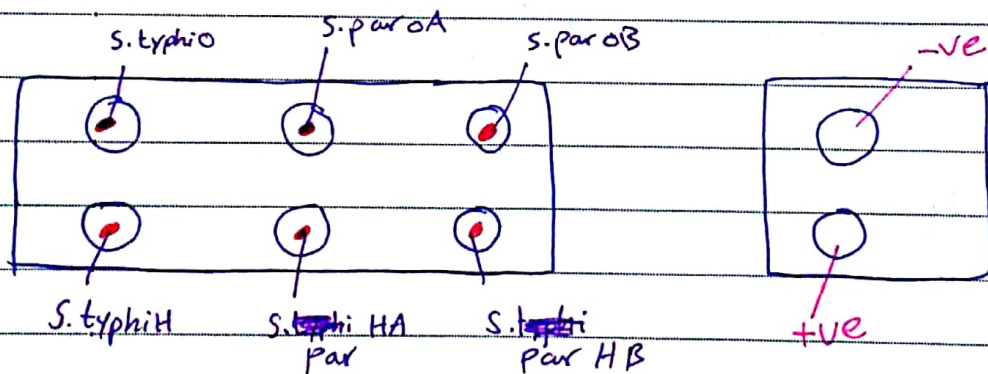
* Titer $\Rightarrow$ Conc. of Ab in blood due to Ag exposed for body or highest dilution that give +ve agg.

* Widal Test $\Rightarrow$ test for Salmonella or S. paratyphi.

Thyphoid fever $\Rightarrow$ due to food poisoning.

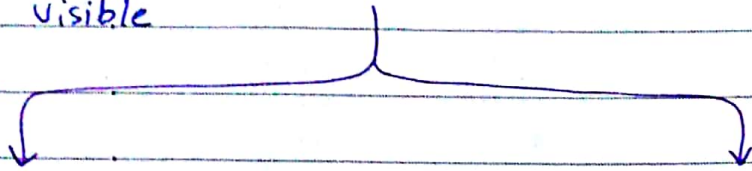
- Somatic Ag $\rightarrow$ O $\rightarrow$ blue
- body Ag.
- Flagella Ag $\rightarrow$ H $\rightarrow$ red

S. typhi O	} $\rightarrow$ blue	S. typhi H	} $\rightarrow$ red
S. par O A		S. par H A	
S. par O B		S. par H B	
x S. Par O C		x S. par H C	



Lab #9 : Passive Agg (Indirect Agg)

* Ag-Ab agg between them need inert substance to make agg visible



Latex agg

Hem agg

* Direct latex agg:-

- Direct X

- looking for Ag

- Indirect

- bind with Ab.

* Indirect latex agg:-

- looking for Ab

- bind with Ag.

* * * *

Latex Agg:-

Application → 1- Detect some hormones.

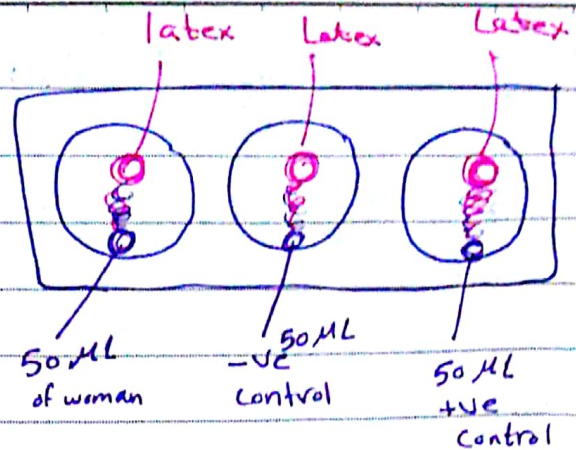
2- " " proteins.

3- " " bacterial infection.

* HCG hormone → Human chorionic gonadotropin hormone.
secreted from placenta of pregnant women in blood first
then in urine after 7-10 days absence of period.

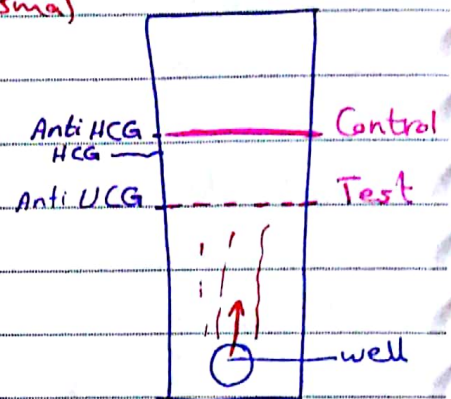
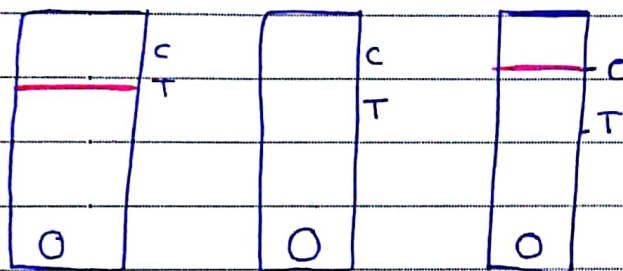
Range 25 IU/L (*)

P.S agg → pregnant
P.S -ve → not pregnant



* long incubation gives false result.

* 2-3 drops (urine, serum, plasma)
* 2 pink lines → pregnant



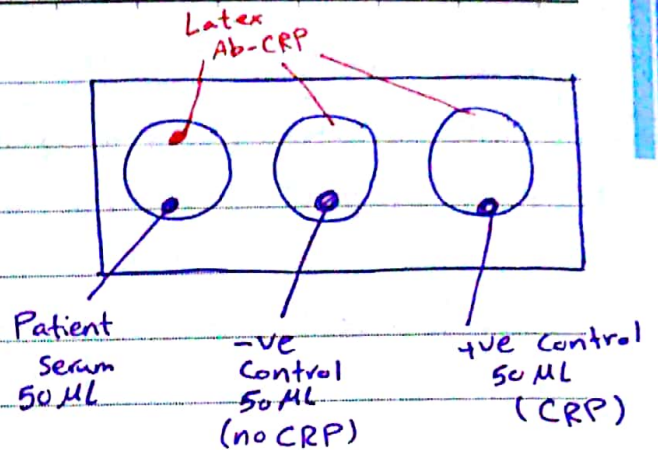
(HCG Pregnancy Test)

* * * * *

CRP → C-reactive protein secreted by hepatocyte upon any infection (Direct Latex Agg).

Range > 0.8 IU/dL

P.S. → +ve infection
P.S. → -ve no infection



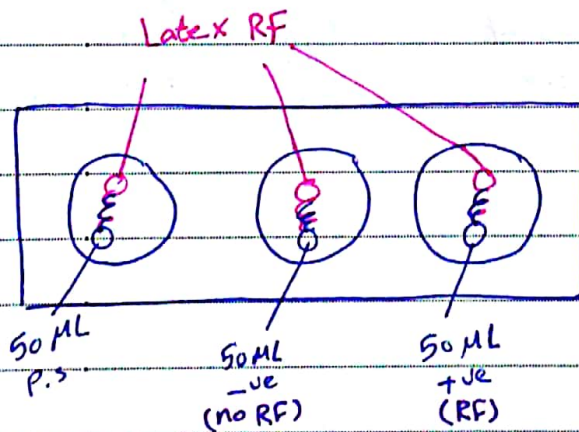
wait 3-5 min / see result.

* * * * *

RF Test → Rheumatoid Factors

- Abs against ~~for~~ F_c portion of Abs.
- Secreted in Rheumatoid fever في الروماتيزم
↳ Rheumatism.

Range 8 IU / L



- * P.S. → +ve Rheumatoid fever
- * P.S. → -ve no Rheumatoid fever.

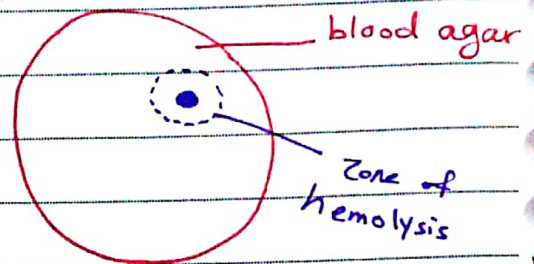
ASO Test:

Anti against streptolysin O enzyme.

S.O enzyme secreted by Streptococcus β -hemolysis.

Through infection $\rightarrow$ Antibiotic

Similarities Ag $\leftarrow$ + Antibody



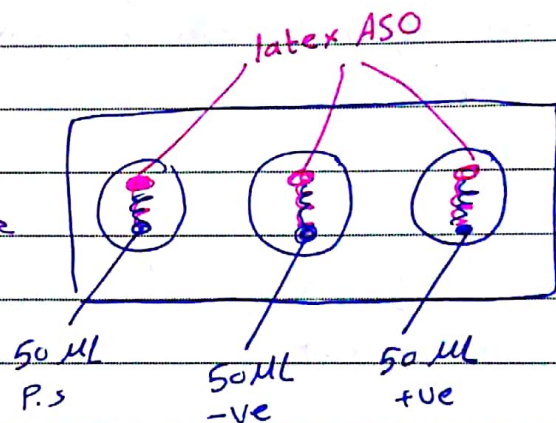
- Joints $\rightarrow$ hematosim.

- ~~reptile~~ kidney $\rightarrow$ nephritis

- Heart $\rightarrow$ Indolcardiatic.

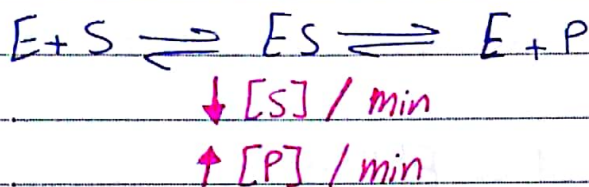
P.S $\rightarrow$ +ve ASO enzyme

P.S $\rightarrow$ -ve no S.O enzyme



Lab #10 :- ELISA

- E → Enzyme
- L → Linked → E-Ab₂ (E-IgG) → Conjugated
- I → Immuno → Ag-Ab rxn
- S → Sorbant → Adsorbed to plastic surface (Ag or Ab)
- A → Assay → 1 unit of time
E-assay



* Characters (Advantages) of ELISA:-

- 1- Simple.
- 2- No expensive tools.
- 3- Safety.
- 4- Sensitive.

* Applications:-

- 1- Tumor marker.
- 2- Some infectious disease (Hb, Ag, AIDS "viruses").
- 3- Some hormones.

* 3 types of wells bottoms:-

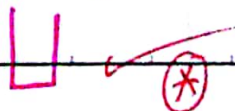
1- V shape.



2- U shape



3- Flat shape



* ELISA → ELISA reader → type of spectrophotometer.

* Types of ELISA:-

1- Sandwich ELISA → "Direct ELISA"

⇒ looking for Ag → Ab linked with plastic.

2- Indirect ELISA

⇒ looking for Ab ⇒ Ag linked plastic.

3- Competitive ELISA → looks like RIA.

* Characters of enzymes used in ELISA:-

① High turnover number.

* ② Can bind Ab₂ (conjugated) E-IgG.

* ③ Enzyme has chromogen substrate.

↳ diff. color. "substrate color, product color"

④ stable (long time).

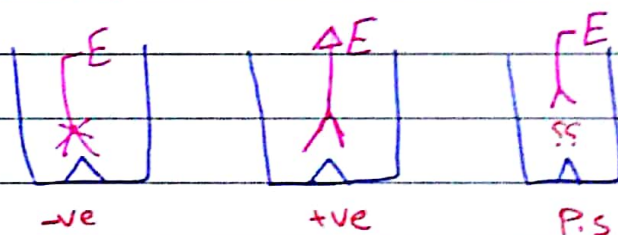
* * * * *

E	Substrate	Product
Alkaline Phosphatase (AP)	PNPP colorless	PNP yellow
Galactosidase	Gla colorless	Gla + substance dark color
Horse radish Peroxidase (HRP)	ABTS colorless →	green
	OPD colorless →	orange
	TMB red →	blue

* Indirect ELISA $\rightarrow$ Hbs Ag

Hbs Ag $\rightarrow$ 50-150 IU/mL

- ① Ag $\rightarrow$ Hbs Ag coated to plastic surface + blocking for nonspecific site.
- ② Add 50 μ L -ve^{control}, 50 μ L +ve control, 50 μ L sample $\rightarrow$ incubation 1 hr + washing.
- ③ Add 50 μ L Conjugated E-IgG $\rightarrow$ incubation for 1 hr.
- ④ Washing 3 or 5 times by buffer
- ⑤ Add 50 μ L substrate chromogen. $\rightarrow$ incubation for 15 min at 37°C
- ⑥ "no washing" Add 50 μ L stop solution (stop E activity) $\rightarrow$ blue
 $\rightarrow$ yellow $\rightarrow$ ELISA reader



* * * * *

Sandwich ELISA:- looking for Ag.

- ① Ab coated plastic surface + blocking for nonspecific binding
- ② Add -ve control, +ve control, P.S $\rightarrow$ incubation + washing.
- ③ Add E-Ab₂ conjugate E-IgG, incubation + washing.
- ④ Add 50 μ L incubate at 37°C (for 30 or 15 min) no washing.
- ⑤ Add stop solution.

* $F \rightarrow$ Flourescent "substance that has colorless, upon exposed to U.v they give visible color.

* Assay $\rightarrow$ product 1 min.

* In F.I.A, we need to use fluorescent microscope.

③ give accurate result.

④ → Qualitative

$\left[\begin{array}{l} \rightarrow +ve \\ \rightarrow -ve \end{array} \right]$ by eye through
fluorescent microscope

↳ by reading O.D by fluorometer attached to flowrescent microscope.

1. Follow histochemical substance. "Chemical of tissue"

for

→ ANA test (Anti Nuclear Antibody).

→ systematic lupus Erythromatosis, Romatoid Arth, Hepatitis
auto immune disease, Autoimmun hemolytic anemia.

3- Detect infection intracellular $\rightarrow$ Virus $\rightarrow$ Respiratory syncytial virus

Bacteria → *Trepanoma pallidum* → caused
syphilis (مرض الزهري)

Parasite $\rightarrow$ Toxoplasma gondii $\rightarrow$ caused
Toxoplasmosis داء القات

Types of fluorescent salt

Flourescence isothiocyanate

FITC

green

Rhodamine isothiocyanate

RI TC

red

F.I.A

Direct F.I.A

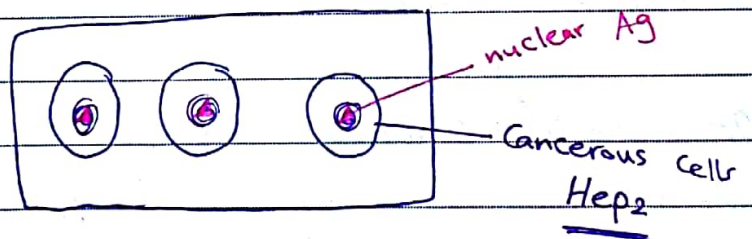
- fluorescent salt attached to Ab_2
- Ab_1 attached to slide

Indirect F.I.A

- looking for Ab_1 , Ag attached to slide.
- fluorescent salt attached to Ab_2 .

ANA :- Indirect F.I.A

- 1- Slide coated with cancerous cells, Hep2 $\rightarrow$ attached to nuclear Ag .



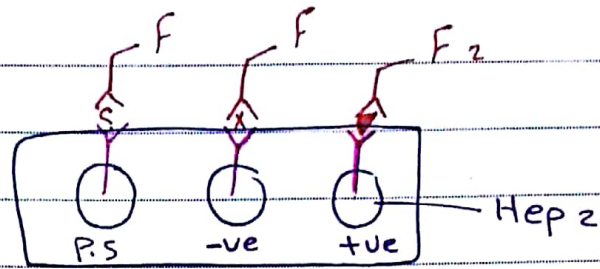
- 2- Add P.S, -ve, +ve control.

incubation $\rightarrow$ washing

- 3- Add F- $Ab_2 \rightarrow$ washing

- 4- Read by ~~fluor~~ fluorescent microscope then O.D by fluorometer.

Direct F.I.A :-



- 1- Slide with Cancerous cells attached by Ab_1 .
- 2- Add P.S, -ve, +ve.
- 3- Add F- $Ab_2 \rightarrow$ washing $[F] \uparrow$, $[Ag] \uparrow$
- 4- Read by flowrosent microscope $\rightarrow$ flowameter.

Lab #12:- R.I.A $\Rightarrow$ Competitive ELISA

* R $\rightarrow$ Radio Radiation come from excited elements (I^{131} , I^{125}) that has isotopes (different in Atomic Mass).

* Advantages:-

- ① very specific
- ② Detect smaller conc. in pg.

* Disadvantages:-

- ① Dangerous substance (radiation).
- ② Cost.
- ③ need specific tool (gamma counter).
- ④ need specific pre-culture.

* Applications:-

- 1- Detect some hormones.
- 2- " tumor marker.
- 3- " Some drugs (Cortisone).

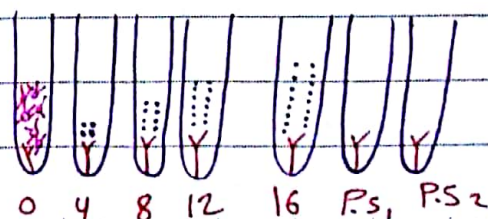
* * * * *

$T_3 \rightarrow$ Thyroxine attached to 3I

$T_4 \rightarrow$ " " " 4I (50-150 IU/mL).

Procedure T_4 :-

① Test tubes attached by fixed conc. of Ab against T_4 .



② Make serial dilution for cold Ag, then add ascending of cold Ag to test tube.

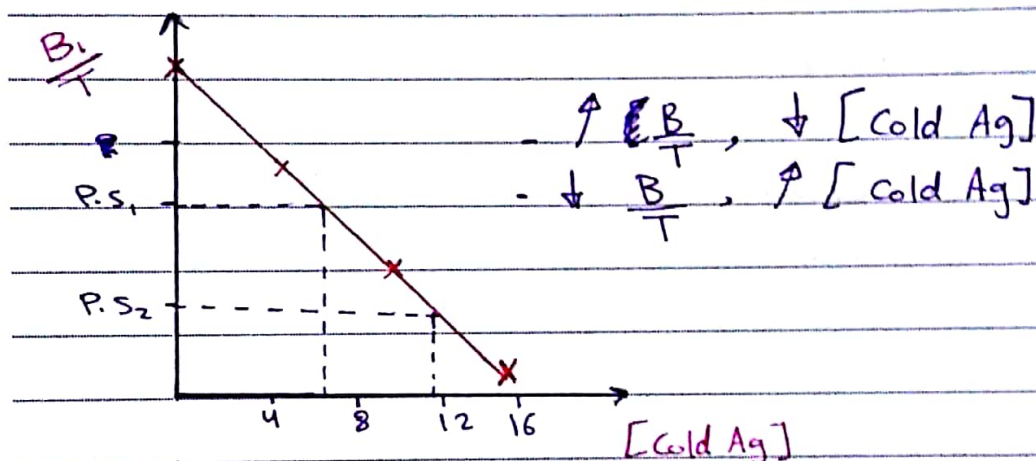
③ Add P.S.1, P.S.2, ...

④ Add fixed conc. of hot Ag 50,000 cpm.

incubation $\rightarrow$ washing.

⑤ Read hot Ag by gamma reader.

⑥ Draw ~~stan~~ standard curve.



I^{125} $\rightarrow$ 60 days
 I^{134} $\rightarrow$ 8 days } $\rightarrow$ half life

The End - ^ ^ 